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Standard Transfer Specification- Compliance test specification- Entity Type A - POSToTokenCarrierInterface application layer protocol for POS devices supporting DKGA=04/EA=11.

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STANDARD TRANSFER SPECIFICATION ASSOCIATION

STANDARD TRANSFER SPECIFICATION –

Compliance Test Specification – Entity Type A - POSToTokenCarrierInterface application layer protocol for POS devices supporting DKGA=04 and EA=11.

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Standard Transfer Specification STS 531-1-0-04 was originally prepared by working group 8.

The text of this standard is based on the following documents:

FDS	Report on voting
STS531-1-0-04/CD	see note1

Note1: due to the large number of documents in the test set, member voting is not performed prior to publication. However, corrections will be made to the document set if errors are reported.

This publication has been drafted in accordance with STSA Directive STS 2100-1 with the exception of Note1

Revision History

Revision	Clause	Date	Change details from previous Edition
1.3	general	August 2015	Initial publication
1.6	general	June 2016	Changed logo
	CTSA01		Updated APDU information, updated tokens to be RND=5
1.6	CTSA15	Nov 2016	Changed SGC to 123458
	CTSA19		Changed $VUDK_3$ and added currency tokens
	CTSA20		Steps 3&4 tokens corrected
	CTSA26		Added 3 KCT test (Step 4)
1.7	General	Nov 2016	Various changes incorporated as per comments received
1.8			Not published
1.8.1	general	Sept 2017	Added VEK classifications to all tests.
	CTSA19		Auto generation of keychange tokens now optional as per IEC62055-41.
	CTSA20-23		Added note regarding maximum transfer amount for currency and limitations of transfer channel
	general		Standardized on all 'Note' entries to normal text
1.8.2		Jan 2018	Updated revision from 1.8.1 to 1.8.2 to match the document set.
1.9	CTSA11	Aug 2017	Added additional Test/Display token values as per 62055-41 Ed3 Table 27
	CTSA16		Added keychange test to an expired KEN and from an expired to a non-expired KEN
	CTSA19		Changed automatic generation of keychange tokens to optional
	CTSA27		Added extended token tests as defined in STS202-5
	CTSA20-23		Added note for maximum transfer amount
	General		Added vending classifications to each test
1.9.1	CTSA14 Step 1	March 2019	Added other utility types.
	CTSA16		Corrected APDU information on KRN
	CTSA26		CTSA26 - changed SGC to 123461 since KEN of 85 for SGC 123460 would not allow the keychange.
	general		Updated annexure A, reformatted document, added foreword
1.9.2	Title		Changed title to remove “optionally” for DKGA04 support
	CTSA20-23 Note		Changed note on currency values supported
	CTSA24-26		Removed first paragraph
	General		Changed note to say that VSM4 and above to be used, or a hardware security module
1.9.3	CTSA01	Jan 2020	Added currency tests.
1.9.4	Foreword	Jan 2021	Added note on voting
1.9.5	CTSA20	April 2022	Changed note on currency range testing
	CTSA16		Steps 3,4 – updated the description of the tests. Changed tokens in Step 4 to use already defined keys in VSM
1.9.6	CTSA01	April 2023	Changed IDRecord to show all steps. Various editorial changes.
2.0	general	Nov 2024	Removed all DKGA02 tests, changed document number from STS531-1-0-04 to STS531-1-11.

			Changed tokens to cater only for Base Dates 2014 and 2035
	4.1		Added note that both 2-digit and 4-digit manufacturer codes must be supported
2.1	general	Nov 2024	Updated document revision to match document set
2.2	CTSA01, CTSA05, CTSA16	July 2025	Corrected tokens

1 Scope

1.1 General

This document provides the compliance criteria and test descriptions for prepayment meters designed to accept tokens that comply with the STS and POS systems designed to produce STS-compliant tokens.

2 Normative references

2.1 General

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 62051 – *ELECTRICITY METERING – Glossary of terms*

IEC 62055-41 – *ELECTRICITY METERING – PAYMENT SYSTEMS – Part 41: Standard Transfer Specification – Application layer protocol for one-way token carrier systems*

STS531-0 *Compliance Test Specification – Quality plan*

STS202-5 - Addendum to IEC62055- 41: Payment Systems - Standard Transfer Specification (STS) - Class 2 token extensions

3 Terms and definitions

3.1 Definitions

For the purposes of this test specification, the definitions given in the normative references identified in paragraph 2 apply.

3.2 Terms

For the purposes of this test specification, the terms given in the normative references identified in paragraph 2 apply.

4 Test requirements and specifications

4.1 Entity type A: POS to Token Carrier Interface – Application layer protocol

4.1.1 General

Each test comprises several steps with associated recordings and expected results. Any deviation from these shall be interpreted as non-compliance and a failure recorded against that step.

Note: every effort has been made to ensure that compliance with the IEC62055-41 standard will be met if all tests in this document are performed successfully. It is however not possible to test all combinations of tokens – it is therefore the manufacturer's responsibility to ensure that the UUT meets all the requirements of IEC62055-41.

Note: All tokens that make use of the random number bits in the 66-bit token field have been generated using an RND value of 5 in this specification.

Note: Virtual Security Module (VSM) Version 6 and above must be used with this specification, with the option to fix RND=5 selected.

Note: Vending classifications V = vending, E = Engineering, and K = Keychange have been added to each test, according to the classifications noted in IEC62055-41 Annex C.3. Systems may be certified according to whatever combination that the manufacturer specifies that his system supports.

Note: All tokens are generated using base dates of 2014 and 2035 only – base date 1993 is no longer tested.

Note: DKGA02 is deprecated and no longer supported.

Note: dates and times are expressed in the following format in this specification: yyyy/mm/dd HH:mm:ss where HH is in 24 hour format.

Note: both 2-digit and 4-digit manufacturer codes must be supported by the POS.

4.1.2 Equipment to be submitted

The following equipment is required for certification:

1. Equipment to be certified that has:
    a. The VUDK's as specified in this document
    b. The capability of generating tokens based on the keys specified in item a above.
    c. The equipment to be certified must be set up to allow the vending of tokens when dates are moved forward and backwards in time from previous tokens. This is to allow for the testing of the different base dates allowed with DKGA=04.

4.1.3 Information to be submitted

Annexure A must be completed by the manufacturer.

4.1.4 CTSA01 – TransferCredit

APDU information to be used for this test:

MeterPAN	600727000000000009, 000001000000000082
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1 (steps 1 – 8), 6 (steps 9-12)
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01) (steps 1-4, 9-12)	600727000000000009===0111201457011
IDRecord (TCT=02) (steps 1-4, 9-12)	600727000000000009===0211201457011
IDRecord (TCT=01) (steps 5-8)	000001000000000082===0111201457011
IDRecord (TCT=02) (steps 5-8)	000001000000000082===0211201457011
Transfer Amount	0.1 kWh (elect), 0.1kl (water), 0.1m^3 (gas), 0.1 min, 100 currency units
TokenIssueDate	see tests
Vending Key VUDK	ABABABABABABABAB9494949494949494012345$67_{16}$
Classification	V

Overview: This test verifies general compliance with respect to the generation of a TransferCredit token.

Electricity Payment Meters – do steps 1,5,9
Water Payment Meters – do steps 2,6,10
Gas Payment Meters – do steps 3,7,11
Time Payment Meters – do steps 4,8,12

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a 0.1kWhTransferCredit token using the information in the APDU above, with a KRN = 1, and a TokenIssueDate of 2014-03-01 13:00:00.	**5862 5789 9869 9135 7486**
2	Generate a 0.1kl TransferCredit token using the information in the APDU above, with a KRN = 1, and a TokenIssueDate of 2014-03-01 13:05:00.	**5113 2789 4367 8786 1315**
3	Generate a 0.1m^3 TransferCredit token using the information in the APDU above, with a KRN = 1, and a TokenIssueDate of 2014-03-01 13:10:00.	**2142 7168 5439 8849 7677**
4	Generate a 0.1min TransferCredit token using the information in the APDU above, with a KRN = 1, and a TokenIssueDate of 2014-03-01 13:15:00.	**5631 5517 0196 4181 8580**
5	Change the MeterPAN to 000001000000000082 and the TokenIssueDate to 2014-03-01 13:40, then generate a 0.1 kWhTransferCredit token, with a KRN = 1.	**1316 5821 0064 0727 9478**
6	Change the MeterPAN to 000001000000000082 and the TokenIssueDate to 2014-03-01 13:45:00. Generate a 0.1 kl TransferCredit token, with a KRN = 1.	**2816 8470 1473 2058 0432**

Step	Instruction	Token data to be transferred to the TCDU
7	Change the MeterPAN to 000001000000000082 and the TokenIssueDate to 2014-03-01 13:50:00. Generate a 0.1 m^3 TransferCredit token, with a KRN = 1.	**1563 2590 8517 9990 5715**
8	Change the MeterPAN to 000001000000000082 and the TokenIssueDate to 2014-03-01 13:55:00. Generate a 0.1min TransferCredit token, KRN = 1.	**5340 4104 5344 4817 3435**
9	Generate a 0.1kWhTransferCredit token using the information in the APDU above with a meter PAN of 600727000000000009, but with a base date of 2035, a KRN of 6, and a TokenIssueDate of 2035-01-01 08:00:00.	**5993 6545 0505 5030 8168**
10	Generate a 0.1kl TransferCredit token using the information in the APDU above with a meter PAN of 600727000000000009, but with a base date of 2035, a KRN of 6, and a TokenIssueDate of 2035-01-01 08:05:00.	**5503 0884 2515 9891 3292**
11	Generate a 0.1m^3 TransferCredit token using the information in the APDU above with a meter PAN of 600727000000000009, but with a base date of 2035, a KRN of 6, and a TokenIssueDate of 2035-01-01 08:10:00.	**1406 4537 3119 1633 7013**
12	Generate a 0.1min TransferCredit token using the information in the APDU above with a meter PAN of 600727000000000009, but with a base date of 2035, a KRN of 6, and a TokenIssueDate of 2035-01-01 08:15:00.	**6462 2162 7715 3760 4642**

4.1.5 CTSA02 – InitiateMeterTest/Display

Overview: This test verifies compliance with respect to the generation of an InitiateMeterTest/Display token (Test 0).

APDU information to be used for this test:

MeterPAN	See test steps
TCT	01, 02
DKGA	<Not specified>
EA	<Not specified>
SGC	<Not specified>
TI	<Not specified>
KRN	<Not specified>
KT	<Not specified>
KeyExpiryNumber	<Not specified>
IDRecord	<Not specified>
Control Field	0xFFFF (65535 decimal)
Cassification	E

Step	Instruction	Token data to be transferred to the TCDU
1	Generate an InitiateMeterTest/Display token using a MeterPAN = 600727000000000009. (11-digit Decoder Reference Number)	**5649 3153 7254 5031 3471**
2	Generate an InitiateMeterTest/Display token using a MeterPAN = 000001000000000082. (13-digit Decoder Reference Number)	**0230 5843 0050 5295 1967**

4.1.6 CTSA03 – SetMaximumPowerLimit

Overview: This test verifies general compliance with respect to the generation of a SetMaximumPowerLimit token.
This test need only be done for electricity Payment Meters.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Transfer Amount	1 kW
Token Date/Time	2024-03-28 09:01
Vending Key VUDK	$\text{ABABABABABABABAB94949494949494940123456 7}_{16}$
Classification	E

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a SetMaximumPowerLimit token using the information in the APDU above.	**3449 4889 4753 7886 6499**

4.1.7 CTSA04 – ClearCredit

Overview: This test verifies general compliance with respect to the generation of a ClearCredit token.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01) (step 1)	600727000000000009===0111201457011
IDRecord (TCT=02) (step 1)	600727000000000009===0211201457011
IDRecord (TCT=01) (step 2)	000001000000000082===0111201457011
IDRecord (TCT=02) (step 2)	000001000000000082===0211201457011
RegisterToClear Val	0XFFFF
Token Date/Time	2024-03-28 09:15
Vending Key VUDK	$ABABABABABABABAB949494949494949401234567_{16}$
Classification	E

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a ClearCredit token using the information in the APDU above.	**1808 7553 1137 0143 9362**
2	Change the MeterPAN to 000001000000000082 and leaving all the other APDU information as in Step1 above, but with a TokenIssueDate of 2016-08-30 09:20:00, generate a ClearCredit token.	**5278 4471 9678 3922 6516**

4.1.8 CTSA05 – Set1stSectionDecoderKey, Set2ndSectionDecoderKey, Set3rdSectionDecoderKey, Set4thSectionDecoderKey

Overview: This test verifies general compliance with respect to the generation of a Keychange token set.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
EA	11
Initial BaseDate	2014/2035
Initial DKGA	04
Initial SGC	201457
Initial TI	01
Initial KRN	1
Initial KT	2
Initial KeyExpiryNumber	255
Initial Rollover Bit	0
Initial IDRecord (TCT=01) (step 1, 2)	600727000000000009===0111201457011
Initial IDRecord (TCT=02) (step 1, 2)	600727000000000009===0211201457011
Initial IDRecord (TCT=02) (step 3)	000001000000000082===0211201457011
New Base Date	2014, 2035 (Step2)
New DKGA	04
New SGC	201457
New TI	02

New KRN	1 (Step1,3),6 (Step2)
New KT	2
New KeyExpiryNumber	255
New Rollover Bit	0
New IDRecord (TCT=01) (step 1, 2)	600727000000000009===0111201457021
New IDRecord (TCT=02) (step 1, 2)	600727000000000009===0211201457021
New IDRecord (TCT=02) (step 3)	000001000000000082===0211201457021
Classification	K

Vending Key	VUDK No	SGC	KRN	KT	KEN
ABABABABABABABAB949494949494949401234567$_{16}$	VUDK$_1$	201457	1	2	255
ABABABABABABABAB949494949494949401234567$_{16}$	VUDK$_2$	201457	6	2	255

Step	Instruction	Tokens to be transferred to the TCDU
1	Generate the Keychange token set using the information in the APDU above with VUDK$_1$.	**0385 4447 0864 5345 1911 (1stKCT)** **0279 1799 4362 4246 1152 (2ndKCT)** **6711 8070 3293 5397 7190 (3rdKCT)** **1901 9128 6909 0293 2218 (4thKCT)**
2	Generate the Keychange token set using the information in the APDU above but change the new base date to 2035, (with VUDK$_2$). New KRN = 6, new SGC = 201457, New TI = 02	**3953 6809 2146 4154 4852 (1stKCT)** **2326 3472 8840 1643 3225 (2ndKCT)** **4526 8691 9859 3933 4940 (3rdKCT)** **6538 9277 4068 8124 2890 (4thKCT)**
3	Change the Meter PAN to 000001000000000082, then generate the Keychange token set using the information in the APDU above (as per Step 1).	**4283 8173 4768 4983 3694 (1stKCT)** **1625 3835 8667 2353 0086 (2ndKCT)** **5337 7655 6515 3668 8781 (3rdKCT)** **1049 8195 1487 7232 0798 (4thKCT)**

4.1.9 CTSA06 – ClearTamperCondition

Overview: This test verifies general compliance with respect to the generation of a ClearTamperCondition token.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Token Date/Time	2024-03-28 10:00
Vending Key VUDK	ABABABABABABABAB949494949494949401234567$_{16}$
Classification	E

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a ClearTamperCondition token using the information in the APDU above.	**0719 4127 7291 9986 2502**

4.1.10 CTSA07 – SetMaximumPhasePowerUnbalanceLimit

Overview: This test verifies general compliance with respect to the generation of a SetMaximumPhasePowerUnbalanceLimit token.
This test need only be done for Electricity Payment Meters.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Token Date/Time	2024-03-28 10:20
MaximumPhasePowerUnbalanceLimit	10 Watts
Vending Key VUDK	$\mathrm{ABABABABABABABAB9494949494949494012345 67}_{16}$
Classification	E

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a SetMaximumPhasePowerUnbalanceLimit token using the information in the APDU above.	**3606 1906 1324 6121 0546**

4.1.11 CTSA08 – RND: RandomNumber

This test has been removed since the vending system has no control over the generation of the random bits. It is included in the compliance test spec for the Security Module.

4.1.12 CTSA09 – TID: TokenIdentifier

Overview: This test verifies general compliance with respect to the TID field, this includes testing for the SpecialReservedTokenIdentifier and multiple tokens generated in the same minute.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
RegisterToClear Val	0XFFFF
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Ist Token Date/Time	2025-03-29 00:00
2nd Token Date/Time	2025-03-29 00:01

3rd Token Date/Time	2025-03-29 00:03
Vending Key VUDK	$ABABABABABABABAB9494949494949494012345 67_{16}$
Classification	V, E
TransferCredit amount	1kWh (elect), 1m^3 (water), 1m^3 (gas), 1min (time), 1credit unit (currency)

For E classification, do only steps 1-3
For V classification, do only steps 4-6
For V and E classification, do either steps 1-3 or 4-6

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a ClearCredit token using the information in the APDU above. (1st token date and time).	**4985 5956 2608 3615 1299**
2	Generate a ClearCredit token using the information in the APDU above. (2nd token date and time).	**6950 2588 4120 1681 4836**
3	Generate three ClearCredit tokens using the information in the APDU above. (3rd token date and time set). These tokens must be generated in the same minute.	The first token in this token set must be: **1444 5046 3619 9349 8655** The second token in this token set must be: **4769 8508 9458 9752 3259** The third token in this token set must be: **6771 4091 9196 2417 4550**
4	Generate a TransferCredit token using the information in the APDU above. (1st token date and time).	**3697 4923 7885 9866 1903 (elect)** **3758 6800 4617 0289 9954 (water)** **3137 8079 9623 3558 6659 (gas)** **1536 8107 5678 3217 6773 (time)** **6726 8527 5006 6103 9337 (elec currency)** **3914 9790 6894 2112 7894 (water currency)** **5303 2483 4281 1003 6122 (gas currency)** **1856 2766 7311 4057 9499 (time currency)**
5	Generate a TransferCredit token using the information in the APDU above. (2nd token date and time).	**1430 1287 8884 1533 8172 (elec)** **6229 1798 8313 8183 4312water)** **5788 8614 9375 3461 2569 (gas)** **3334 2440 7036 3985 3579 (time)** **2271 9196 9310 9755 0751 (elec currency)** **4948 2305 9401 5379 9703 (water currency)** **4404 9498 9402 0236 3494 (gas currency)** **2950 2969 0383 0204 0648 (time currency)**

6	Generate three TransferCredit tokens using the information in the APDU above. (3rd token date and time set). These tokens must be generated in the same minute.	**3845 7689 5329 1702 49233167 6902 5713 8366 9696 (elec)** **1215 6402 0245 6047 2125** **2609 0142 3533 4747 2322** **3373 7445 6648 5129 0941 (water)** **6018 6206 7015 0999 6547** **7071 7004 6289 3165 2568** **0385 5631 0148 4907 4718 (gas)** **1515 8997 6109 4696 2756** **0700 2062 9385 2859 5759** **1151 8273 9819 7073 7359 (time)** **2320 4323 1036 1042 2176** **4350 1303 5131 7991 1562** **1014 0453 7766 7031 6809 (elec currency)** **4721 5165 5050 0257 2266** **4795 6056 3460 5563 7960** **6700 8743 7066 2763 1105 (water currency)** **1263 0685 0045 6946 4531** **0936 2002 2419 9175 9208** **0019 8824 0976 3229 3970 (gas currency)** **3374 1802 8533 8251 5887** **2406 1141 0969 5854 7614** **1672 5733 3255 0759 9265 (time currency)** **5332 8471 5314 8974 4368**

4.1.13 CTSA10 – Amount: TransferAmount

Overview: This test verifies general compliance with respect to the calculation of the TransferAmount field.

Note: if a full set of test steps have been performed for any payment meter type (electricity, water, gas or time), it is not necessary to repeat the test steps in any of the other payment meter types.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Vending Key VUDK	ABABABABABABABAB9494949494949494901234567$_{16}$
Classification	V

Table 1 – Transfer Amount Table

Test Number	TokenIssueDate	TransferAmount
1	2026-07-11 00:30	25.6 kWh, kl, m3, min
2	2026-07-11 00:35	1638.3 kWh, kl, m^3, min
3	2026-07-11 00:40	1638.4 kWh, kl, m^3, min
4	2026-07-11 00:45	2000.0 kWh, kl, m^3, min
5	2026-07-11 00:50	18022.3 kWh, kl, m^3, min
6	2026-07-11 00:55	18022.4 kWh, kl, m^3, min
7	2026-07-11 01:44	181862.3 kWh, kl, m^3, min
8	2026-07-11 02:49	181862.4 kWh, kl, m^3, min
9	2026-07-11 03:54	1820162.4 kWh, kl, m^3, min

Step	Instruction	Token data to be transferred to the TCDU
1	Generate a TransferCredit token using the information given in Table 1. above for test number 1 if steps 10-18 for water or Steps 19-27 for gas or steps 28-36 for time have not been done.	**2364 1553 6653 9815 4702**
2	As above for test number 2.	**2589 1865 0802 0361 7683**
3	As above for test number 3.	**2761 5769 0590 1211 0139**
4	As above for test number 4.	**6633 4093 5690 9327 5152**
5	As above for test number 5.	**5446 9257 3741 5949 6590**
6	As above for test number 6.	**3215 2659 9491 3797 0265**
7	As above for test number 7.	**5173 7434 3056 3514 4582**
8	As above for test number 8.	**0370 4036 2700 7495 5413**
9	As above for test number 9.	**2758 9639 2484 6008 7544**
10	Perform test steps 10-18 for water TransferCredit tokens if Steps 1-9 for electricity or Steps 19-27 for gas or steps 28-36 for time have not been done As above for test number 1.	**0958 3268 1976 4671 7908**
11	As above for test number 2.	**1470 1698 7704 3089 6760**
12	As above for test number 3.	**3400 2938 9228 4926 6884**
13	As above for test number 4.	**6714 2506 8204 1267 3779**
14	As above for test number 5.	**3887 9081 9196 6507 1085**
15	As above for test number 6.	**7351 0229 7603 7122 4859**
16	As above for test number 7.	**1289 2756 0126 4422 3549**

Step	Instruction	Token data to be transferred to the TCDU
17	As above for test number 8.	**3908 0250 4476 7958 7151**
18	As above for test number 9.	**1862 4152 5952 0954 4039**
19	Perform test steps 19-27 for gas TransferCredit tokens if Steps 1-9 for electricity or Steps 10-18 for water or steps 28-36 for time have not been done. As above for test number 1.	**4498 6114 0898 6617 2191**
20	As above for test number 2.	**2220 1321 7510 0536 7123**
21	As above for test number 3.	**6603 2647 6049 0606 2472**
22	As above for test number 4.	**0040 2401 1490 1042 6708**
23	As above for test number 5.	**3055 1311 4855 5861 2857**
24	As above for test number 6.	**6529 4529 7081 1735 6090**
25	As above for test number 7.	**3949 0034 2888 4949 5633**
26	As above for test number 8.	**0492 1144 6489 6657 6098**
27	As above for test number 9.	**1661 2725 7175 2225 1986**
28	Perform test steps 28-36 for time TransferCredit tokens if Steps 1-9 for electricity or Steps 10-18 for water or steps 19-27 for gas have not been done. As above for test number 1.	**4374 5038 1145 6187 9026**
29	As above for test number 2.	**5510 9344 9987 4486 7787**
30	As above for test number 3.	**3831 7498 7680 7287 0648**
31	As above for test number 4.	**5106 7046 8792 4003 1527**
32	As above for test number 5.	**1595 5566 4041 9214 2866**
33	As above for test number 6.	**2008 2638 7702 0338 1988**
34	As above for test number 7.	**1770 4294 2977 2493 9274**
35	As above for test number 8.	**2554 9082 0239 2627 9371**
36	As above for test number 9.	**1642 6719 0471 9318 8319**

4.1.14 CTSA11 – Control: InitiateMeterTest/DisplayControlField

Overview: This test verifies compliance with respect to the generation of an InitiateMeterTest/Display token, the InitiateMeterTest/DisplayControlField with respect to its calculation and bit position in the token is also verified.

APDU information to be used for this test:

MeterPAN	<Not specified>
TCT	01,02
DKGA	<Not specified>
EA	<Not specified>
SGC	<Not specified>
TI	<Not specified>
KRN	<Not specified>
KT	<Not specified>
KeyExpiryNumber	<Not specified>
IDRecord	<Not specified>
Classification	E

Note: the first token shown in each step below is for 2-digit manufacturer codes, and the second token shown is for 4 digit manufacturer codes.

Step	Instruction	Token to be transferred to the TCDU
1	Generate an InitiateMeterTest/Display token using the information given above and an InitiateMeterTest/DisplayControlField value of 000000001_{16}.	**0000 0000 0001 5099 7584** **0115 2921 5090 3605 4672**
2	As above, with a DisplayControlField value of 000000002_{16}.	**0000 0000 0001 6777 4880** **0115 2921 5133 3104 2448**
3	As above, with a DisplayControlField value of 000000004_{16}.	**0000 0000 0002 0132 8896** **0115 2921 5219 2095 2465**
4	As above, with a DisplayControlField value of 000000008_{16}.	**1844 6744 0738 4377 2416** **0115 2921 5391 0083 8034**
5	As above, with a DisplayControlField value of 000000010_{16}.	**3689 3488 1475 5332 2496** **0115 2921 5734 6054 3637**
6	As above, with a DisplayControlField value of 000000020_{16}.	**0000 0000 0006 7109 3248** **0115 2921 6421 8002 0378**
7	As above, with a DisplayControlField value of 000000040_{16}.	**0000 0000 0012 0797 4400** **0115 2921 7796 1897 3828**
8	As above, with a DisplayControlField value of 000000080_{16}.	**0000 0000 0022 8172 8512** **0115 2922 0544 9688 0824**
9	As above, with a DisplayControlField value of 000000100_{16}.	**0000 0000 0044 2920 8064** **0115 2922 6042 5269 4700**

Step	Instruction	Token to be transferred to the TCDU
10	As above, with a DisplayControlField value of 000000200_{16}.	**0000 0000 0087 2419 5840** **0115 2923 7037 6432 2536**
11	As above, with a DisplayControlField value of 000000400_{16}.	**0000 0000 0173 1410 5857** **0115 2925 9027 8757 7952**
12	As above, with a DisplayControlField value of 000002000_{16}.	**0000 0000 1375 7317 3770** **0115 2956 6891 1315 4192**
13	As above, with a DisplayControlField value of 000004000_{16}.	**0000 0000 2750 1212 7252** **0115 2991 8734 8524 9680**
14	As above, with a DisplayControlField value of 000008000_{16}.	**0000 0000 5498 9003 4216** **0115 3062 2422 2942 8368**
15	As above, with a DisplayControlField value of 000010000_{16}.	**0000 0001 0996 4584 8124** **0115 3202 9797 1778 8816**
16	As above, with a DisplayControlField value of 000020000_{16}.	**0000 0002 1991 5747 5960** **0115 3484 4546 9451 4832**

4.1.15 CTSA12 – MPL: MaximumPowerLimit

Overview: This test verifies general compliance with respect to the calculation of the MaximumPowerLimit field.

Note: this test only required if the payment meter is an electricity payment meter.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Vending Key VUDK	$\text{ABABABABABABABAB9494949494949494}01234567_{16}$
Classification	E

Step Number	TokenIssueDate	MaximumPowerLimit
1	2027-12-15 07:00	256 watts
2	2027-12-15 07:05	16383 watts
3	2027-12-15 07:10	16384 watts
4	2027-12-15 07:15	20000 watts
5	2027-12-15 07:20	180223 watts
6	2027-12-15 07:25	180224 watts
7	2027-12-15 07:30	1818623 watts

8	2027-12-15 07:35	1818624 watts
9	2027-12-15 07:40	18201624 watts

Step	Instruction	Token to be transferred to the TCDU
1	Generate a SetMaximumPowerLimit token using the information given above for step number 1.	**0869 6877 1074 8977 2342**
2	As above for step number 2.	**1298 1641 7617 7049 0283**
3	As above for step number 3.	**7046 6640 4584 9784 4971**
4	As above for step number 4.	**0609 6117 3895 0988 1002**
5	As above for step number 5.	**2022 0432 6213 9191 3192**
6	As above for step number 6.	**1821 3277 1304 0981 9867**
7	As above for step number 7.	**6359 1892 7560 6959 8223**
8	As above for step number 8.	**0850 2056 5962 6768 7018**
9	As above for step number 9.	**4681 0404 5146 2659 1449**

4.1.16 CTSA13 – MPUL: MaximumPhasePowerUnbalanceLimit

Overview: This test verifies general compliance with respect to the calculation of the MaximumPhasePowerUnbalanceLimit field.

Note: this test only required if the payment meter is an electricity payment meter.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Vending Key VUDK	$\text{ABABABABABABABAB94949494949494940123456}7_{16}$
Classification	E

Notes:

- The dates and times together with the MaximumPhasePowerUnbalanceLimit values to be used are specified in the table below.

Step Number	TokenIssueDate	MaximumPhasePowerUnbalanceLimit
1	2027-12-15 08:00	256 watts
2	2027-12-15 08:05	16383 watts
3	2027-12-15 08:10	16384 watts
4	2027-12-15 08:15	20000 watts
5	2027-12-15 08:20	180223 watts
6	2027-12-15 08:25	180224 watts
7	2027-12-15 08:30	1818623 watts
8	2027-12-15 08:35	1818624 watts
9	2027-12-15 08:40	18201624 watts

Step	Instruction	Token to be transferred to the TCDU
1	Generate a SetMaximumPhasePowerUnbalanceLimit token using the information given above for step number 1.	**3365 9574 9387 0204 9734**
2	As above for step number 2.	**3786 5901 1754 6043 5734**
3	As above for step number 3.	**1494 5501 1906 3537 0267**
4	As above for step number 4.	**6658 7780 6874 6200 0679**
5	As above for step number 5.	**3952 5526 5094 2578 5956**
6	As above for step number 6.	**1330 6442 6388 2810 7790**
7	As above for step number 7.	**0322 3893 0485 5078 6553**
8	As above for step number 8.	**3905 9189 0299 9562 5564**
9	As above for step number 9.	**1167 8265 9824 6226 9502**

4.1.17 CTSA14 – Register: RegisterToClear

Overview: This test verifies general compliance with respect to the calculation of the RegisterToClear field.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Vending Key VUDK	ABABABABABABABAB9494949494949494012345677$_{16}$
Classification	E

Notes:

- The dates and times together with the RegisterToClear values to be used are specified in the table below. i.e. register 0000$_{16}$ requires a datetime of 2034-07-03 09:00, and register 0003$_{16}$ requires a datetime of 2034-07-03 09:03
- Step 1 - it is only necessary to do the utility supported
- Steps 3 to 6 need only be done if the UUT supports the generation of CurrencyTokens.

Step Number	TokenIssueDate	RegisterToClear
1	2034-07-03 09:00 2034-07-03 09:01 2034-07-03 09:02 2034-07-03 09:03	0000$_{16}$ (elec) 0001$_{16}$ (water) 0002$_{16}$ (gas) 0003$_{16}$ (time)
2	2034-07-03 09:05	FFFF$_{16}$
3	2034-07-03 09:10	0004$_{16}$
4	2034-07-03 09:15	0005$_{16}$
5	2034-07-03 09:20	0006$_{16}$
6	2034-07-03 09:25	0007$_{16}$

Step	Instruction	Token to be transferred to the TCDU
1	Generate a ClearCredit token using the information given above for step number 1. Do only the utility supported.	**5080 1913 3186 4462 6174 (Elect)** **3583 6988 0587 5379 3602 (Water)** **5526 5472 2168 4538 6252 (Gas)** **6610 2829 9128 1151 1687 (Time)**
2	As above for step number 2.	**4364 0653 4192 0264 3137**
3	As above for step number 3. See note above.	**2734 1708 3515 1923 1651**
4	As above for step number 4. See note above.	**0264 1726 2132 6253 7268**
5	As above for step number 5. See note above.	**5788 8853 6112 5166 2352**
6	As above for step number 6. See note above.	**7140 6825 2986 9260 2735**

4.1.18 CTSA15 – NKHO: NewKeyHighOrder, NKLO: NewKeyLowOrder, NKMO1: NewKeyMidOrder1, NKMO2: NewKeyMidOrder2, KENHO: KeyExpiryNumberHighOrder, KENLO: KeyExpiryNumberLowOrder, SGCLO: SGCLowOrder, SGCHO: SGCHighOrder, RO: RolloverKeyChange

Overview: This test verifies general compliance with respect to the generation of a Set1stSectionDecoderKey token and a Set2ndSectionDecoderKey token.

APDU information to be used for this test:

MeterPAN	600727111111111153
TCT	02
EA	11
Initial VUDK	$\text{ABABABABABABABAB9494949494949494012345 67}_{16}$
Initial DKGA	04
Initial Base Date	2014
Initial SGC	201457
Initial TI	01
Initial KRN	1
Initial KT	2
Initial KeyExpiryNumber	255
Initial IDRecord (TCT=01)	600727111111111153===0111201457011
Initial IDRecord (TCT=02)	600727111111111153===0211201457011
New VUDK	$\text{ABABABABABABABAB949494949494949401234567}_{16}$
New Base Date	2035
New DKGA	04
New SGC	201457
New TI	02
New KRN	6
New KT	2
New KeyExpiryNumber	255
New IDRecord (TCT=01)	600727111111111153===0111201457021
New IDRecord (TCT=02)	600727111111111153===0211201457021
Classification	K

Step	Instruction	Tokens to be transferred to the TCDU
1	Generate the Keychange token set using the information in the APDU above.	**3664 4361 0984 1237 4610 (1stKCT)** **7073 9203 5541 4636 9967 (2ndKCT)** **6250 2430 2732 5646 3906 (3rdKCT)** **0146 7733 7655 2609 5667 (4thKCT)**

4.1.19 CTSA16 – KeyExpiryNumber

Overview: This test verifies general compliance with respect to the generation of a Credit token and Keychange token set where the KeyExpiryNumber has expired.

APDU information to be used for this test:

MeterPAN	600727111111111153
TCT	01, 02
EA	11
VUDK	$\text{ABABABABABABABAB949494949494949401234567}_{16}$
DKGA	04
SGC	201460
TI	01
KRN	7
KT	2
Base Date	2014
KeyExpiryNumber	85
IDRecord (TCT=01)	600727111111111153===0111201457011
IDRecord (TCT=02)	600727111111111153===0211201457011

Token Date/Time	2032/10/27 08:00 (date after KEN value)
Classification	V, E, K
Transfer amount	1kWh (elect), 1m^3 (water), 1m^3 (gas), 1min (time) 1credit unit (currency)

For classification V, do step 1 only
For classification E, do step 2 only
For classification K, do steps 3 and 4 only

Step	Instruction	Expected Result
1	Generate a Transfer Credit Token using the information in the APDU above.	The POS should indicate a key expiry. No token should be generated.
2	Generate a ClearAllCredit Token using the information in the APDU above.	The POS should indicate a key expiry error. No token should be generated.
3	Generate the Keychange token set using the information in the APDU above. Initial SGC = 201460 Initial KEN = 85 (0X55) Initial KRN = 7 Initial TI = 01 New SGC = 201460 New KEN = 85 (0x55) New KRN = 7 New TI = 02	The POS should indicate a key expiry error. No tokens should be generated.
4	Generate the Keychange token set using the information in the APDU above. Initial SGC = 201460 Initial KEN = 85 (0X55) Initial KRN = 7 Initial TI = 01 New SGC = 201461 New KEN = 255 (0xFF) New KRN = 8 New TI = 01	**4245 9678 1148 4470 2446 (1stKCT)** **6952 7545 3808 2004 6989 (2ndKCT)** **0473 4524 2287 2638 7550 (3rdKCT)** **3343 4270 8219 4146 1228 (4thKCT)**

4.1.20 CTSA17 – DRN Check Digit

Overview: This test verifies general compliance with respect to the verification of the Check Digit in the DRN.

Classification: V, E, K

Step	Instruction	Expected Result
1	Enter the following DRNs (Meter Numbers) into the POS **12345678904** **11111111113**	The POS should indicate a Luhn digit error in the DRN number in both cases.

4.1.21 CTSA18 – DateOfExpiry

Overview: This test verifies general compliance with respect to the verification of a meter ID card that implements date of expiry. This test is only valid for POS devices that support this feature.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	02
EA	11
SGC	201457
TI	01
KRN	1
DOE	06/2024
IDRecord (TCT=01)	600727000000000009=2406=0111201457011
IDRecord (TCT=02)	600727000000000009=2406=0211201457011
Classification	V, E, K

Step	Instruction	Expected Result
1	Generate a meter ID card with the IDRecord as shown above.	
2	Read the meter Id card with the POS device whose date is set at any date after 01/07/2024.	The POS device should indicate that the ID card has expired.

4.1.22 CTSA19 – Automatic generation of Set1stSectionDecoderKey, Set2ndSectionDecoderKey, Set3rdSectionDecoderKey, Set4thSectionDecoderKey

Overview: This test verifies general compliance with respect to the automatic generation of a Set1stSectionDecoderKey token, a Set2ndSectionDecoderKey, a Set3rdSectionDecoderKey, and a Set4thSectionDecoderKey token.

Note: It is only necessary to perform tests for one meter type (electricity, water, time, or gas), and only if the manufacturer has stated in Table 2 that the automatic generation of keychange tokens is supported.

Note: The process for these tests is as follows -

1. The relevant attribute is changed (TI, KRN, KEN, SGC) for the meter - this may be done directly in the meter table in the database, or by some other method in the POS.
2. This results in a "pending keychange" required for the meter - no keychange tokens are generated at this time.
3. When a credit token is next generated, the POS will detect this attribute has changed for the meter, generate the keychange tokens (since one of the above attributes have changed), and then the credit token encrypted with the new key.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
Base Date	2014
EA	11
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0111201457011
IDRecord (TCT=02)	600727000000000009===0211201457011
Transfer Amount	0.1 kWh, 0.1 kl, 0.1 m^3, 0.1 min, 0.1-unit currency tokens
Token Date/Time	2024-04-06 10:00
Classification	V & K

Vending Key	VUDK No	SGC	KRN	KT	KEN
ABABABABABABABAB949494949494949401234567$_{16}$	VUDK$_1$	201457	1	2	255
ABABABABABABABAB949494949494949401234567$_{16}$	VUDK$_2$	201457	2	2	255
ABABABABABABABAB949494949494949401234567$_{16}$	VUDK$_2$	201457	4	2	170
ABABABABABABABAB949494949494949401234567$_{16}$	VUDK$_3$	123461	8	2	255

Step	Instruction	Expected Result
1	Change the Tariff Index associated with the DRN from ‘01’ to ‘02’, and the token issue date to 2024-04-06 10:00. Then, using the new APDU information, with $VUDK_1$ generate a TransferCredit token. Initial TI = 1 New TI = 2 Initial KRN = 1 New KRN = 1 Initial SGC = 201457 New SGC = 201457 Initial VK = $VUDK_1$ (with KEN = 255) New VK = $VUDK_1$ (with KEN = 255)	The following keychange tokens shall be generated: **0385 4447 0864 5345 1911 (1stKCT)** **0279 1799 4362 4246 1152 (2ndKCT)** **6711 8070 3293 5397 7190 (3rdKCT)** **1901 9128 6909 0293 2218 (4thKCT)** As well as one of the following credit tokens. **1809 6548 2776 2437 1750 (Electricity)** **5045 8697 5050 2563 6371 (Water)** **6436 3177 1915 9381 9185 (Gas)** **6055 5173 4046 9011 7861 (Time)** **5470 1429 7190 7006 4827 (Elec Currency)** **2698 2859 1854 5387 4280 (Water Currency)** **0644 1367 3951 9312 9146 (Gas Currency)** **0580 6081 7077 9730 1660 (Time Currency)**

2	Change the KRN from 1 to 2 and change the token issue date to 2024-04-06 10:10. Then, using the new APDU information, with $VUDK_2$, generate a TransferCredit token. Initial TI = 1 New TI = 1 Initial KRN = 1 New KRN = 2 Initial SGC = 201457 New SGC = 201457 Initial VK = $VUDK_1$ (with KEN = 255) New VK = $VUDK_2$ (with KEN = 255)	The following keychange tokens shall be generated: **2654 6541 1568 6342 7612 (1stKCT)** **6798 0973 7533 0708 0757 (2ndKCT)** **5201 2045 0042 1440 0808 (3rdKCT)** **3879 8311 1021 0542 9368 (4thKCT)** As well as one of the following credit tokens. **4687 2490 6835 7358 0920 (Electricity)** **5663 6169 8295 8548 5383 (Water)** **2866 0992 3382 9361 2898 (Gas)** **0035 8750 1217 6691 4749 (Time)** **2812 6004 0570 8838 9705 (Elec Currency)** **7168 8257 0544 2318 6101 (Water Currency)** **4365 2832 2094 2636 4749 (Gas Currency)** **6597 5218 0031 3895 5507 (Time Currency)**
3	Change the KeyExpiryNumber associated with the SGC from '255' to '170', and the token issue date to 2024-04-06 10:15. Then, using the new APDU information, with $VUDK_2$, generate a TransferCredit token. Initial TI = 1 New TI = 1 Initial KRN = 2 New KRN = 4 Initial SGC = 201457 New SGC = 201457 Initial VK = $VUDK_2$ (with KEN = 255) New VK = $VUDK_2$ (with KEN = 170)	The following keychange tokens shall be generated: **6694 9074 4429 3352 9273 (1stKCT)** **0737 7215 8503 7536 8570 (2ndKCT)** **2547 8881 0903 0699 9193 (3rdKCT)** **2903 5779 1068 6385 0442 (4thKCT)** As well as one of the following credit tokens. **1160 3655 8246 4526 4466 (Electricity)** **2749 1131 1390 6069 7331 (Water)** **1405 9864 0948 1683 7230 (Gas)** **6413 7242 6580 3846 8332 (Time)** **5869 7732 1581 3713 1455 (Elec Currency)** **1668 7266 2487 1447 8995 (Water Currency)** **6834 8525 5281 3847 9610 (Gas Currency)** **0142 2998 9605 2659 5722 (Time Currency)**

4	Change the SGC associated with the DRN from '201457' to '201461', and the token issue date to 2024-04-06 10:20. Then, using the new APDU information, with $VUDK_3$, generate a TransferCredit token. Initial TI = 1 New TI = 1 Initial KRN = 1 New KRN = 8 Initial SGC = 201457 New SGC = 201461 Initial VK = $VUDK_1$ (with KEN = 255) New VK = $VUDK_3$ (with KEN = 255)	The following keychange tokens shall be generated: **6009 0865 0801 1326 1203 (1stKCT)** **6137 6774 7001 5399 9017 (2ndKCT)** **0102 7794 4505 3511 4909 (3rdKCT)** **5738 3055 0701 9634 1904 (4thKCT)** As well as one of the following credit tokens. **3201 7959 9118 8292 0368 (Electricity)** **2345 7780 6068 3428 7972 (Water)** **5582 0018 6294 3270 6891 (Gas)** **3281 8632 4595 5262 3177 (Time)** **0372 2498 3304 3039 6550 (Elec Currency)** **3982 1890 5107 9585 1783 (Water Currency)** **4639 6827 1760 4577 4190 (Gas Currency)** **7237 4710 7493 3021 0638 (Time Currency)**

Note:

CTSA20- CTSA23 - In the case where the system cannot resolve the transfer amount to the range of the transfer amount as specified in IEC62055-41, then test to the range that the system is able to resolve, as specified in the submitted documentation.

4.1.23 CTSA20 – Currency Token (Electricity Credit)

This test shall only be performed if the manufacturer has indicated that the UUT supports the generation of Electricity TransferCredit (Currency) tokens, otherwise this test may be omitted.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
Vending Key VUDK	ABABABABABABABAB949494949494949401234567$_{16}$
Classification	V

Step Number	TokenIssueDate	TransferAmount
1	2027-08-21 12:01:00	1
2	2027-08-21 12:02:00	16383
3	2027-08-21 12:03:00	16384
4	2027-08-21 12:04:00	180224
5	2027-08-21 12:05:00	1818624
6	2027-08-21 12:06:00	18202624
7	2027-08-21 12:07:00	182042624
8	2027-08-21 13:03:00	1820442624
9	2027-08-21 13:04:00	18204442624
10	2027-08-21 13:05:00	1.82044E+14
11	2027-08-21 13:10:00	1.82044E+20
12	2027-08-21 13:11:00	1.82044E+29
13	2027-08-21 13:12:00	-1
14	2027-08-21 13:14:00	-180224
15	2027-08-21 13:15:00	-1.82044E+14

Overview: This test verifies general compliance with respect to the generation of a TransferCurrency token, and tests exponent value calculation.

Step	Instruction	Expected Result
1 - 15	Generate TransferCurrency tokens with the information in the APDU above, and for each of the steps indicated above.	The token for each step must be identical to the token corresponding to the step number specified in the token table below.

Token Table	
Step	**Token Decimal Digits**
1	**7250 3171 5014 6492 5869**
2	**5250 5389 0637 5815 9292**
3	**3653 5899 4982 5029 6222**
4	**6731 6201 5786 4924 0766**

5	**2910 0783 7496 5687 8839**
6	**2495 9219 4719 6603 6989**
7	**1788 7120 2192 1303 5288**
8	**3618 8384 2056 6492 9478**
9	**6894 4298 4882 3910 4097**
10	**5059 6393 7549 5546 9432**
11	**0314 5904 2928 5937 6346**
12	**4074 1014 0130 3472 5556**
13	**4015 6929 5960 1947 8780**
14	**6775 1816 0553 7582 3218**
15	**6169 8173 4580 8228 4427**

4.1.24 CTSA21 – Currency Token (Water Credit)

This test shall only be performed if the manufacturer has indicated that the UUT supports the generation of Water TransferCredit (Currency) tokens, otherwise this test may be omitted.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	02
DKGA	04
BaseDate	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
Vending Key VUDK	$ABABABABABABABAB9494949494949494\allowbreak 01234567_{16}$
Classification	V

Step Number	TokenIssueDate	TransferAmount
1	2028-05-13 10:01:00	1
2	2028-05-13 10:02:00	16383
3	2028-05-13 10:03:00	16384
4	2028-05-13 10:04:00	180224
5	2028-05-13 11:00:00	1818624
6	2028-05-13 11:01:00	18202624
7	2028-05-13 11:02:00	182042624
8	2028-05-13 11:03:00	1820442624
9	2028-05-13 11:04:00	18204442624
10	2028-05-13 11:05:00	1.82044E+14
11	2028-05-13 11:06:00	1.82044E+20
12	2028-05-13 11:07:00	1.82044E+29
13	2028-05-13 11:12:00	-1
14	2028-05-13 11:13:00	-180224
15	2028-05-13 11:14:00	-1.82044E+14

Overview: This test verifies general compliance with respect to the generation of a TransferCurrency token, and tests exponent value calculation.

Step	Instruction	Expected Result
1 - 15	Generate TransferCurrency tokens with the information in the APDU above, and for each of the steps indicated above.	The token for each step must be identical to the token corresponding to the step number specified in the token table below.

Token Table	
Step	**Token Decimal Digits**
1	**5306 3466 3805 9666 1701**
2	**5986 3900 5197 8738 3729**
3	**1755 4659 2541 6892 2816**
4	**7027 5770 1230 0067 4568**
5	**1095 0477 4779 7145 8096**
6	**1848 9392 8751 0032 1647**
7	**7307 6026 4457 1326 5347**
8	**2707 5722 4203 0009 7008**
9	**6118 7598 5096 8166 1359**
10	**0499 4536 7887 6195 9275**
11	**4129 6967 9586 4029 7949**
12	**6835 0070 4923 5488 7884**
13	**6643 4263 9858 9194 3927**
14	**3169 3163 7056 1349 6038**
15	**5661 2752 5791 8666 5500**

4.1.25 CTSA22 – Currency Token (Gas Credit)

This test shall only be performed if the manufacturer has indicated that the UUT supports the generation of Gas TransferCredit (Currency) tokens, otherwise this test may be omitted.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	02
DKGA	04
Base Date	2014
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
Vending Key VUDK	$\text{ABABABABABABABAB9494949494949494012345 67}_{16}$
Classification	V

Step Number	**TokenIssueDate**	**TransferAmount**
1	2025-05-22 10:01:00	1
2	2025-05-22 10:02:00	16383
3	2025-05-22 10:03:00	16384
4	2026-05-21 10:59:00	180224
5	2026-05-21 11:00:00	1818624
6	2026-05-21 11:01:00	18202624
7	2026-05-21 11:02:00	182042624
8	2026-05-21 11:03:00	1820442624
9	2026-05-21 11:04:00	18204442624
10	2026-05-21 11:05:00	1.82044E+14
11	2026-05-21 11:06:00	1.82044E+20
12	2026-05-21 11:07:00	1.82044E+29
13	2026-05-21 11:08:00	-1
14	2026-05-21 11:09:00	-180224
15	2026-05-21 11:10:00	-1.82044E+14

Overview: This test verifies general compliance with respect to the generation of a TransferCurrency token, and tests exponent value calculation.

Step	Instruction	Expected Result
1 - 15	Generate TransferCurrency tokens with the information in the APDU above, and for each of the steps indicated above.	The token for each step must be identical to the token corresponding to the step number specified in the token table below.

Token Table	
Step	**Token Decimal Digits**
1	**0148 6360 3516 9985 3267**
2	**2872 3971 3078 1626 9512**
3	**2296 2026 8647 6028 6738**
4	**6711 4805 4871 3895 9112**
5	**3926 4667 4421 1396 9134**
6	**2083 3061 0520 0835 0952**
7	**1101 3442 6386 0647 5263**
8	**0470 6102 2495 9954 6605**
9	**4436 8728 0254 5759 9399**
10	**5424 0591 6064 6354 3746**
11	**5980 3276 5919 2123 6711**
12	**0185 3361 1066 6790 3886**
13	**5527 8980 3095 5916 5499**
14	**3756 2956 7177 1621 5775**
15	**3087 3765 9772 9343 5624**

4.1.26 CTSA23 – Currency Token (Time Credit)

This test shall only be performed if the manufacturer has indicated that the UUT supports the generation of Time TransferCredit (Currency) tokens, otherwise this test may be omitted.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	02
DKGA	04
Base Date	1993
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
Vending Key VUDK	$ABABABABABABABAB94949494949494940123456 7_{16}$
Classification	V

Step Number	TokenIssueDate	TransferAmount
1	2027-05-22 21:01:00	1
2	2027-05-22 21:02:00	16383
3	2028-05-21 23:03:00	16384
4	2028-05-21 23:04:00	180224
5	2028-05-21 23:05:00	1818624
6	2028-05-21 23:06:00	18202624
7	2028-05-21 23:07:00	182042624

8	2028-05-21 23:08:00	1820442624
9	2028-05-21 23:09:00	18204442624
10	2028-05-21 23:10:00	1.82044E+14
11	2028-05-21 23:11:00	1.82044E+20
12	2028-05-21 23:12:00	1.82044E+29
13	2028-05-21 23:13:00	-1
14	2028-05-21 23:14:00	-180224
15	2028-05-21 23:15:00	-1.82044E+14

Overview: This test verifies general compliance with respect to the generation of a TransferCurrency token, and tests exponent value calculation.

Step	Instruction	Expected Result
1 - 15	Generate TransferCurrency tokens with the information in the APDU above, and for each of the steps indicated above.	The token for each step must be identical to the token corresponding to the step number specified in token table 1 below.

Token Table 1	
Step	**Token Decimal Digits**
1	**5567 4953 0547 8600 6863**
2	**3246 5783 9938 2212 7585**
3	**6048 3348 5608 1594 9040**
4	**4063 4728 8045 8794 4982**
5	**5053 9387 1581 0924 8658**
6	**6056 3588 5331 2124 2212**
7	**5927 4692 0908 0016 5946**
8	**7340 6516 8803 5711 2594**
9	**2940 5441 1725 7753 2316**
10	**6865 7003 7481 3306 4593**
11	**0803 1464 3915 3904 0624**
12	**4132 3680 7461 2663 5691**
13	**4527 8828 3296 5725 0646**
14	**1918 1847 6589 5016 7811**
15	**6427 3920 9239 9234 1792**

4.1.27 CTSA24 – Extended token set

These tests shall only be performed if the manufacturer has indicated that the UUT supports the generation of the extended token set in Class2 Subclass10 as specified in STS202-5. Otherwise, these tests may be omitted.

APDU information to be used for this test:

MeterPAN	600727000000000009
TCT	01, 02
DKGA	04
EA	11
SGC	201457
TI	01
KRN	1
KT	2
KeyExpiryNumber	255
IDRecord (TCT=01)	600727000000000009===0107201457011
IDRecord (TCT=02)	600727000000000009===0207201457011
Token Issue Date	Per test steps
Vending Key VUDK	ABABABABABABABAB949494949494949401234567$_{16}$
Base Date	2014
Classification	E

Step	Instruction	Tokens to be transferred to the TCDU
1	Using the information in the APDU above withToken date/Time of 2029-05-21 09:00:00, generate a Class2 SubClass10 token. (Index = 63, FlagIndex = 0, FlagValue = 0)	**0717 1716 7159 6451 2818**
2	Using the information in the APDU above withToken date/Time of 2029-05-21 09:05:00, generate a Class2 SubClass10 token. (Index = 63, FlagIndex = 0, FlagValue = 1)	**5812 5830 6458 6653 1183**
3	Using the information in the APDU above withToken date/Time of 2029-05-21 09:10:00, generate a Class2 SubClass10 token. (Index = 0, ControlValue = 0)	**7224 4336 1024 6632 9056**
4	Using the information in the APDU above withToken date/Time of 2029-05-21 09:15:00, generate a Class2 SubClass10 token. (Index = 0, ControlValue = 0123)	**2747 3808 8426 1992 2147**
5	Generate a Class1 Subclass 2 ReadControl token (control value = 0x0)	**0230 5843 0093 4791 4912**
6	Generate a Class1 Subclass 2 ReadFlag token (control value = 0xFC0000000)	**0344 0750 1154 4527 9822**

5 Annexure A – Compliance Verification Request

1.	Manufacturer:	
2.	Product Name/Model:	
3.	Product Software Version:	
4.	Manufacturer's Contact:	
5.	Date:	

Table 2 - Entity Type A Supplier Submitted Information

1.	TCTs supported	01	02	(Tick what is applicable)	
2.	Is Date Of Expiry Supported	Yes	No		
3.	Is STS202-5 supported	Yes	No		
4.	Is automatic generation of keychange tokens supported	Yes	No		
5.	Vending System Classification	V (vending)	E (Engineering)	K (Keychange)	
6.	Manufacturer confirms that Class2 reserved for proprietary use tokens are NOT available on the POS.	Signed:			
7.	Manufacturer confirms that the POS is enabled to generate tokens for all STS manufacturer numbers, and that there is no possibility of excluding any manufacturer number.	Signed:			
8.	State which Utility is Supported (if the system supports units-based meters)	Electricity	Water	Gas	Time
9.	State which Currency is Supported (if the system supports currency-based meters)	Electricity	Water	Gas	Time
10.	State the maximum transfer amount supported in currency tokens (leave blank if the full range is supported)				

6 Annexure B – Test overviews

All tests test for the general token acceptance and PM display indicators, token bit values, amount calculations, CRC, and others, as well indirectly testing for the implementations indicated in the comments column.

Test No	Description	IEC62055-41 ED3 Applicable Clause	Other implementations tested
CTSA01	TransferCredit	6.2, 6.3, 6.4, 6.5	
CTSA02	InitiateMeterTest/Display	6.2, 6.3, 6.4	
CTSA03	SetMaximumPowerLimit	6.2, 6.3, 6.4, 6.5	
CTSA04	ClearCredit	6.2, 6.3, 6.4, 6.5	
CTSA05	Keychange	6.2, 6.3, 6.4, 6.5	
CTSA06	ClearTamperCondition	6.2, 6.3, 6.4, 6.5	
CTSA07	SetMaximumPhasePowerUnbalanceLimit	6.2, 6.3, 6.4, 6.5	
CTSA09	TID: TokenIdentifier	6.2, 6.3, 6.4, 6.5	
CTSA10	TransferAmount	6.2, 6.3, 6.4, 6.5	
CTSA11	InitiateMeterTest/DisplayControlField	6.2, 6.3, 6.4	
CTSA12	MaxiumumPowerLimit	6.2, 6.3, 6.4, 6.5	
CTSA13	MaxiumumPhasePowerUnbalanceLimit	6.2, 6.3, 6.4, 6.5	
CTSA14	RegisterToClear	6.2, 6.3, 6.4, 6.5	
CTSA16	KeyExpiryNumber	6.1, 6.3, 6.5	
CTSA17	DRN Check Digit	6.1.2.3.4	
CTSA18	DateOfExpiry	6.1.11	
CTSA19	Automatic generation of KCT	6.5.2.1	
CTSA20 - CTSA23	CurrencyToken	6.3.2.1, 6.3.2.2, 6.3.6.3, 6.3.22	
CTSA24	Extended token set	STS202-5	